

Homework 4

Actuarial Corporate Finance.

1. $E[R_A] = 1.12$ $E[R_B] = 1.15$ $w_A = 0.6$ $w_B = 0.4$

σ_A σ_B $r_f = 0.05$

$\sigma_A = 0.25$ $\sigma_B = 0.40$

$\text{Cov}(A, B) = 0$

$r_f = 0.05$

Sharpe Ratio: $\frac{E[R_{PF}] - r_f}{\sigma_{PF}}$

$\text{Var}(R_{PF}) = w_i^2 \text{Var}(R_i) + w_j^2 \text{Var}(R_j) + 2w_i w_j \rho_{ij} \sigma_i \sigma_j$

$E[R_{PF}] = \mu = \sum w_i E[R_i]$

$E[R_{PF}]$

$\Rightarrow E[R_{PF}] = \mu_{PAB} = 0.6 \times 0.12 + 0.4 \times 0.15 = 0.132$

$\Rightarrow \mu_{PAB} = 13.2\%$

$\text{Var}(R_{PAB}) = 0.6^2 (0.25)^2 + 0.4^2 (0.4)^2 + 0 \times [\cdot]$

$= 0.0481 \Rightarrow \sigma_{PAB} = 0.2193 \Rightarrow \sigma_{PAB} = 21.93\%$

$\Rightarrow \text{Sharpe Ratio: } \frac{0.132 - 0.05}{0.2193} = 0.373917$

$\text{Sharpe Ratio} = 37.4\%$

2. $\rho_{AB} = 0.8$ $r_A = 0.10$ $r_B = 0.20$ $\sigma_A = 0.2$ $\sigma_B = 0.5$ $w_A = 0.5 = w_B$ $r_f = 0.02$

$\rho_{AC} = 0.1$ $\rho_{BC} = 0.4$ $w_A = 0.5$ $w_B = 0.5$ $\sigma_C = 0.3$

Formula: $r_f + \frac{\sigma_C}{\sigma_P} \rho_{PC} (E[r_P] - r_f) < r_C$

$\sigma_P = \sqrt{0.5^2 \times 0.2^2 + 0.5^2 \times 0.5^2} = 0.2693$ $r_P = 0.5 \times 0.1 + 0.5 \times 0.2 = 0.15$

$\rho_{xy} = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y} \Rightarrow \rho_{AB} = \frac{\text{Cov}(r_A, r_B)}{0.2 \times 0.5} = 0.8 \Rightarrow \text{Cov}(r_A, r_B) = 0.08$

$0.1 = \frac{\text{Cov}(r_A, r_C)}{0.2 \times 0.3} \Rightarrow \text{Cov}(r_A, r_C) = 0.06$

$0.4 = \frac{\text{Cov}(r_B, r_C)}{0.5 \times 0.3} \Rightarrow \text{Cov}(r_B, r_C) = 0.06$

$\Rightarrow \text{Cov}(r_C, 0.5r_A + 0.5r_B) = 0.5\text{Cov}(r_A, r_C) + 0.5\text{Cov}(r_B, r_C) = 0.5 \times 0.06 + 0.5 \times 0.06 = 0.03$

$\Rightarrow \rho_{PC} = \frac{0.03}{0.3 \times 0.2693} = 0.40847$

$\Rightarrow r_C \geq 0.02 + \left(\frac{0.3}{0.2693} \right) (0.40847) (0.15 - 0.02) = 0.07915$